Unit 7 Quiz: Probability Distributions

Name _____

Show your work.

7.1 - Probability Distributions

1. **I check if a probability distribution is valid, and I draw it as a graph.**

   (a) Is this table a valid probability distribution? Explain.

x	1	2	3	4
$P(x)$	0.2	0.5	0.4	-0.1

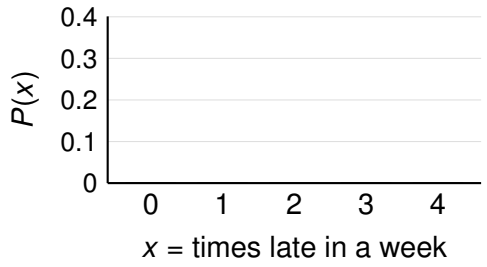
Let X = the number of times the school bus is late in one week.

x	0	1	2	3	4
$P(x)$	0.35	0.30	?	0.10	0.05

   (b) Find the missing probability, $P(2)$.

   (c) Find $P(X \geq 2)$.

   (d) Draw a bar graph of the distribution.



2. **I explain what a random variable is, and I make theoretical and empirical distributions.**

   A box has 10 cards. 5 cards show the number 1. 3 cards show 2. 2 cards show 5. You draw one card without looking. Let X = the number on the card.

   (a) Why is X a random variable?

   (b) Make the **theoretical** probability distribution of X.

x	1	2	5
$P(x)$			

   Ms. Diaz's class drew a card 50 times. They put the card back each time. They got a 1 on 22 draws, a 2 on 18 draws, and a 5 on 10 draws.

   (c) Make the **empirical** probability distribution of X.

x	1	2	5
$P(x)$			

   (d) Why are the two distributions not the same?

7.2 - Expected Value

3. **I find the expected value of a random variable, and I explain what it means.**

A basketball team scores 0, 1, 2, or 3 points each time it has the ball. Let X = the number of points it scores on one turn with the ball.

x (points)	0	1	2	3
$P(x)$	0.45	0.05	0.35	0.15
$x \cdot P(x)$				

(a) Fill in the last row of the table.

(b) Find $E(X)$.

(c) Which statement best describes the meaning of $E(X)$? Circle the letter.

A The team scores exactly 1.2 points each time it has the ball.
B Over many turns with the ball, the team scores about 1.2 points per turn, on average.
C The team scores 1 point more often than any other number of points.
D Over many turns with the ball, the team scores 1.2 points in total.

(d) The team has the ball 60 times in a game. About how many points do you expect it to score?

4. **I decide if a method of choosing is fair, and I decide if a game is fair.**

Lin, Omar, and Sara need to choose who presents first.

(a) Which method is fair? Circle the letter.

A Roll a number cube. 1 or 2: Lin. 3: Omar. 4, 5, or 6: Sara.
B Roll a number cube. 1 or 6: Lin. 2 or 5: Omar. 3 or 4: Sara.
C Flip a coin 3 times. 3 heads: Lin. 3 tails: Omar. Anything else: Sara.
D Spin a spinner with 5 equal parts. 2 parts: Lin. 2 parts: Omar. 1 part: Sara.

(b) Explain your choice.

At a school fair game booth, you roll a number cube. You win $4 if you roll a 1 or a 2. You pay $3 for any other number.

(c) Find the expected value of what you win.

(d) Is the game fair? Explain.

(e) How much should you win for a 1 or a 2 to make the game fair?

7.3 - Expected Payoff

5. **I find the net gain for each outcome, and I find the expected payoff of a lottery, game, or plan.**

   The soccer team sells 100 raffle tickets for $5 each. One ticket wins $200. Five tickets win $20 each. The rest win nothing.

   (a) Fill in the table.

Outcome	Net gain x	$P(x)$	$x \cdot P(x)$
Win $200			
Win $20			
Win nothing			

   (b) Find the expected payoff.

   (c) What does the expected payoff mean?

   (d) The team sells all 100 tickets. How much money does the team raise?

6. **I use expected value, and risk, to make a decision.**

   A tablet screen costs $180 to fix. The chance that a screen breaks in one year is 0.15. A one-year protection plan costs $35.

   (a) Find the expected cost to fix the screen with no plan.

   (b) Which is cheaper on average: buying the plan or not? By how much?

   (c) Give a reason someone might still buy the plan.

   (d) The company sells 2,000 of these plans. About how much money does it expect to make?